

Module 8 - Arithmetic Operators

1. What are Operators?

Operators are symbols or keywords used to perform operations on values and variables, such as arithmetic, relational (comparison), assignment, and logical operations.

2. What are Operands?

- Operands are the values or variables on which an operator performs an operation.
- **Example:** In $10 + 20$, 10 and 20 are operands, and + is the operator.

3. What are Arithmetic Operators?

Arithmetic Operators are used to perform mathematical operations on values and variables, such as addition, subtraction, multiplication, division, modulus, exponentiation, increment, and decrement.

Example Code:

```
// Arithmetic Operators

console.log(10 + 20); // Addition ==> 30

console.log(20 - 5); // Subtraction ==> 15

console.log(20 * 2); // Multiplication ==> 40

console.log(20 / 2); // Division - Quotient ==> 10

console.log(21 % 5); // Modulus - Remainder ==> 1

console.log(3 ** 4); // Exponentiation ==> 81


// Increment ( ++ )

let num = 15;

num++; // Post Increment ==> num = num + 1 ==> 15 + 1 = 16

++num; // Pre Increment ==> num = num + 1 ==> 16 + 1 = 17

console.log(num); // 17


// Decrement ( -- )

let num1 = 20;

num1--; // Post Decrement ==> num1 = num1 - 1 ==> 20 - 1 = 19

--num1; // Pre Decrement ==> num1 = num1 - 1 ==> 19 - 1 = 18

console.log(num1); // 18
```

- **Addition (+)** – Used to add two or more values.
- **Subtraction (-)** – Used to subtract one value from another.
- **Multiplication (*)** – Used to multiply two or more values.
- **Division (/)** – Used to divide one value by another and get the quotient.
- **Modulus (%)** – Used to find the remainder after division.
- **Exponentiation (**)** – Used to raise a number to the power of another number.
- **Increment (++)** – Used to increase the value of a variable by 1.
- **Decrement (--)** – Used to decrease the value of a variable by 1.